Problem Based Learning Session 3

PHAS0050 : Climate & Energy

Prof. Chris Howard

- Nuclear continued
- Fuel Utilisation
- Conversion Ratio
- Moderation
- European Pressurised Reactor

Question 1

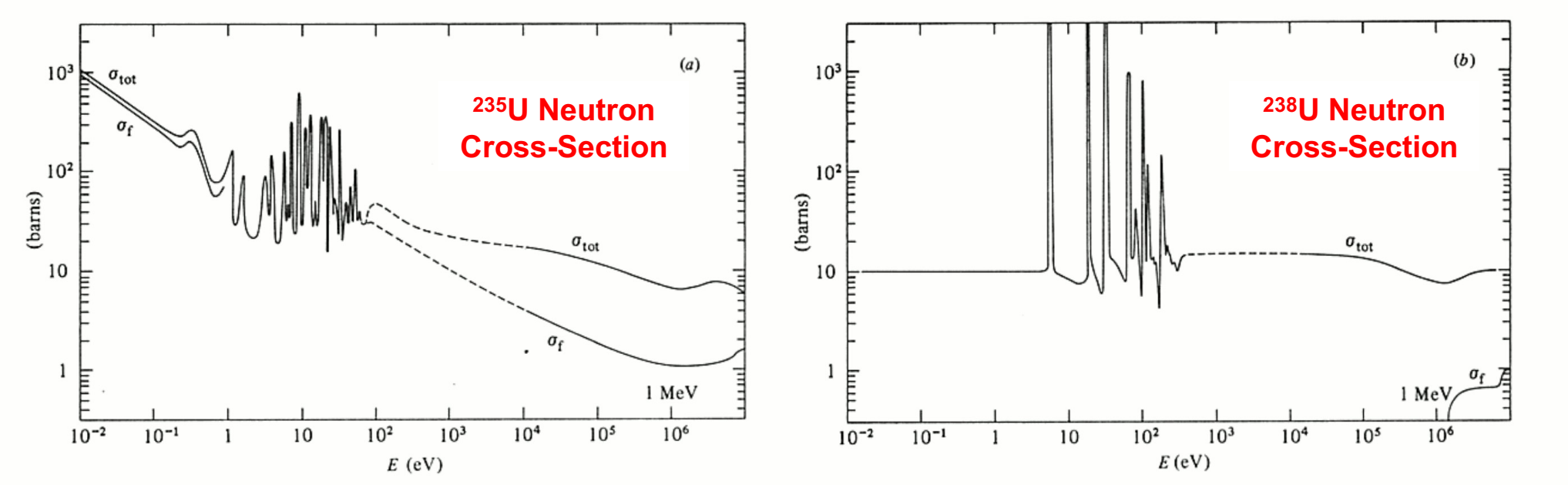
- What is the energy yield per tonne of heavy metal fuel? (fuel burnup)`
- What is the waste volume?

<u>Fuel burnup of EPR :</u>

<u>Waste volume :</u>

Neutron Reactions

Natural uranium: 99.3% ^{238}U + 0.7% ^{235}U

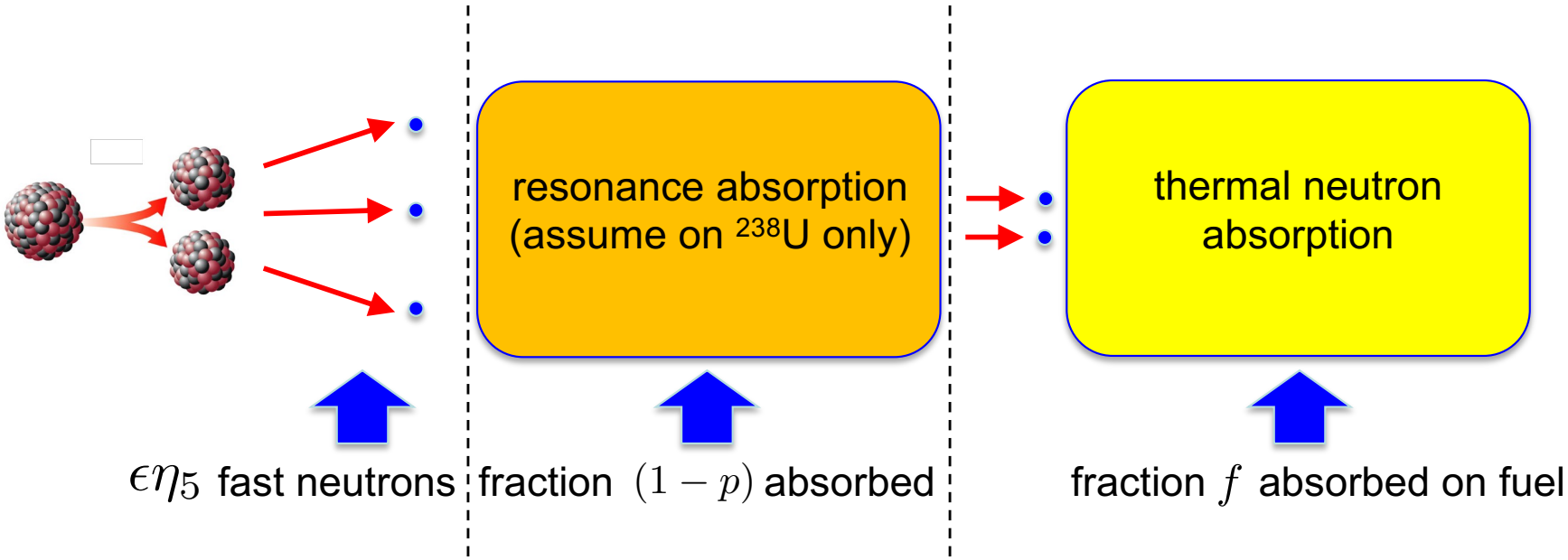


Important features of the different energy regimes :

- Low energy (E_n<0.1 eV) : $\sigma^{235} \gg \sigma^{238}$; mainly fission for ^{235}U
- Resonance region (0.1 eV < E_n< 1 keV) : mainly fission (^{235}U) ; capture (^{238}U)
- Intermediate energies (1 keV < E_n< 1 MeV) : similar σ's . Elastic/inelastic scattering .
- High energies : (E_n> 1 MeV) : ^{238}U becomes fissionable . $\sigma^{235} \sim \sigma^{238}$

Conversion Ratio

- Recall that neutron capture on ^{238}U produces fissile ^{239}Pu
- How many neutrons captured on ^{238}U per neutron absorbed on a ^{235}U nucleus ?



<u>Number of fast neutrons absorbed on ^{238}U :</u>

Conversion Ratio

Number of thermal neutrons absorbed on ^{238}U :

Total conversion ratio :

$$C = \eta_5 \epsilon (1 - p) + \left(\frac{\eta_5}{\eta} - 1 \right)$$

Fuel Utilisation

- Consider the complete burn-up of N fissile nuclei :
  - CN further fissile nuclei created. If they are completely consumed then :
    - C^2N further fissile nuclei created …
- Therefore the total number of nuclei consumed is equal to :

$$N \times (1 + C + C^2 + ...) = \frac{N}{1 - C}$$

- In practice, the total fraction of nuclei consumed in a fuel is given by :

$$\boxed{y(\%) = \frac{x_{\text{initial}}^{\text{fissile}}(\%) - x_{\text{final}}^{\text{fissile}}(\%)}{1 - C}}$$

- C is now an effective (“operational” conversion ratio).

Fuel Utilisation : EPR

Example of EPR reactor with the following parameters :

$$\eta_5 = 2.05,\ \eta = 1.9,\ p = 0.88,\ \epsilon = 1.05$$

$$x^{235}_{\text{initial}} = 5\%,\ x^{\text{fissile}}_{\text{final}} = 1.5\%$$

Initial conversion ratio :

Fuel utilisation with an operational conversion ratio C= 0.42 :

Question 1

- What is the energy yield per tonne of heavy metal fuel? (fuel burnup)`
- What is the waste volume?

<u>Fuel burnup of EPR :</u>

<u>Waste volume :</u>

Moderators

- General considerations :
  - Small mass number A to maximise neutron energy loss through scattering.
  - Small neutron absorption cross-section.

$$\frac{\langle \Delta \ln(E) \rangle \times \sigma_s}{\sigma_c}$$

Material	A	σ_c (barns)	**Figure of Merit**
H	1	0.33	-
D	2	0.0005	-
H_2O	-	-	71
D_2O	-	-	5670
C	12	0.0035	192

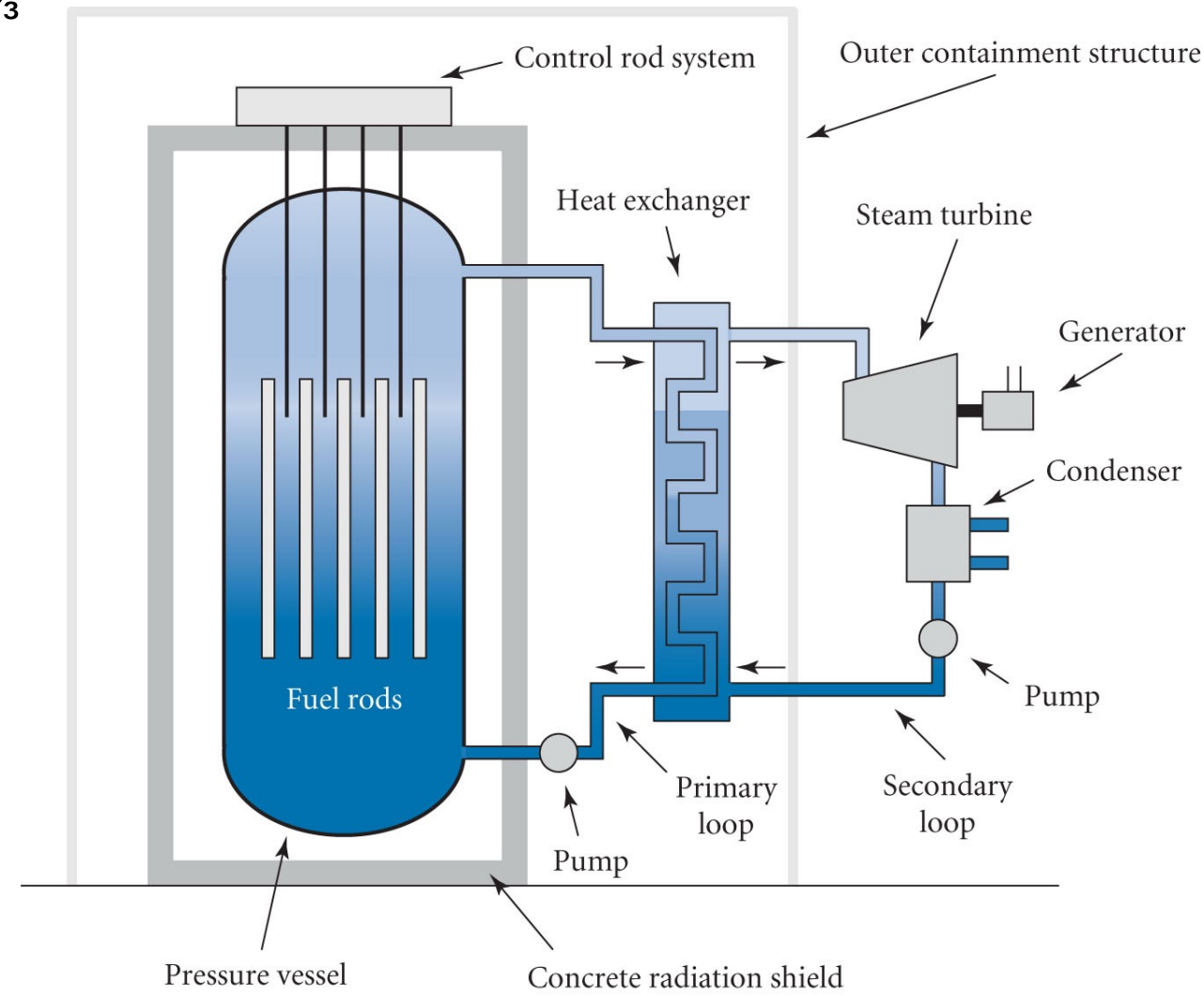
Matrix of *Thermal* Reactor Types

		Moderator		
		H_2O	D_2O	Carbon
Coolant	H_2O	PWR BWR (EPR)	Advanced CANDU Reactor	RBMK (former Soviet Union)
	D_2O		CANDU	
	CO_2			(Advanced) Gas Cooled Reactor
	Helium			High Temperature Gas Cooled Reactor

Pressurised Water Reactor (PWR)

PWR

- Most common reactor type (~⅔ of current world fleet).
- Water kept in liquid phase under high pressure (155 bar); operates up to 325°C.
- Heat-exchanger/steam-generator; secondary circuit drives turbines.
- Fuel (enriched UO_2 clad in Zircaloy) contained in pellets inside rods.
- Control rods lowered into position.
- *Negative void coefficient* (moderation strongly reduced when steam bubbles form).



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2016 Q4

The critical point of water occurs at a temperature of 374 °C and a pressure of 218 bar. Explain how this critical point limits the performance of a Pressurised Water Reactor and estimate the maximum Carnot efficiency of such reactors. (Absolute zero is −273.15 °C) **[3]**

AGR : T_{max} = 640 °C

VHTR : T_{max} = 1000 °C

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- More realistic maximum temperature (for an EPR reactor)

- Core outlet:
  - BE = 330.0°C
  - TH = 331.6°C
  - ME = 327.6°C.

  (warm dome configuration)

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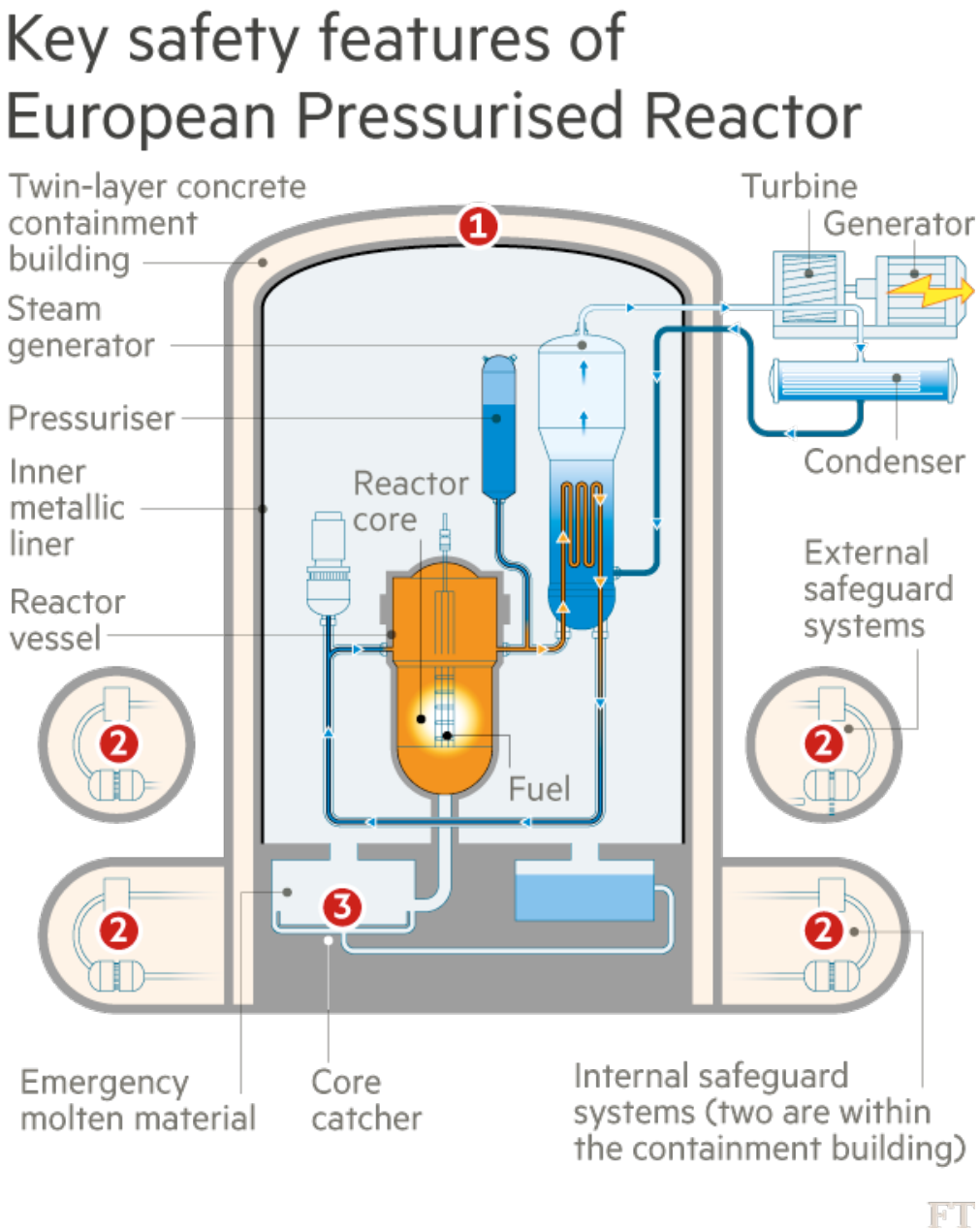
Suppose a fleet of EPR reactors each with a capacity of 1600 MW_e are constructed to supply 50% of the UK's current electricity generating requirement of 45 GW_e. If the estimated core damage frequency of the EPR reactors is 1.0×10^{-6} per reactor per year, what is the chance of a core damage event across the fleet during 50 years of operation? **[2]**

Briefly explain why core damage accidents will not in general lead to a large release of radioactivity into the environment. **[2]**

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